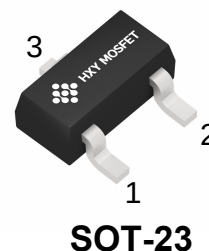




产品描述

XC6206P332MR是具有高纹波抑制率、低功耗、低压差，具有过流和短路保护的CMOS降压型电压稳压器。这些器件具有很低的静态偏置电流（3.0μA Typ.），它们能在输入、输出电压差极小的情况下提供300mA的输出电流，并且仍能保持良好的调整率。由于输入输出间的电压差很小和静态偏置电流很小，这些器件特别适用于希望延长电池寿命的电池供电类产品，如计算机、消费类产品和工业设备等。



管脚编号	管脚名	功能描述
1	VSS	接地
2	VOU	输出
3	VIN	电源输入

产品特点

- ◇ 最大输出电流：300mA
- ◇ 压差电压：160mV@50mA
- ◇ 最大工作电压：8V
- ◇ 输出电压范围：3.3V
- ◇ 高精度：2%
- ◇ 极低的静态工作电流：3μA (典型值)
- ◇ 内置过流和短路保护电路
- ◇ 工作温度范围：-40℃ ~ 85℃

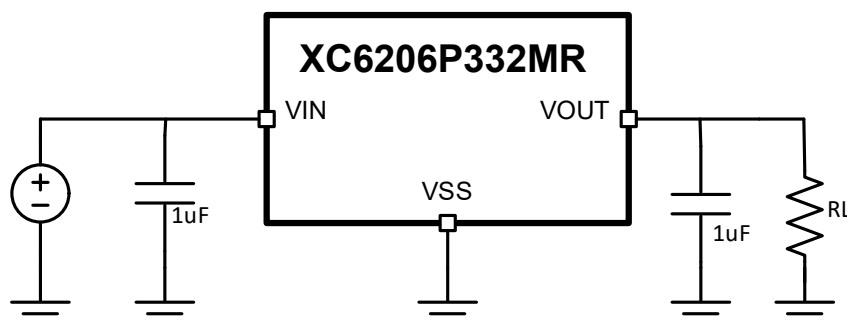
应用领域

- ◇ 电池供电系统
- ◇ 无绳电话设备
- ◇ 无线控制系统
- ◇ 便携/手掌式计算机
- ◇ 便携式消费类设备
- ◇ 便携式仪器
- ◇ 汽车电子设备
- ◇ 电压基准源

应包装标识与订购信息

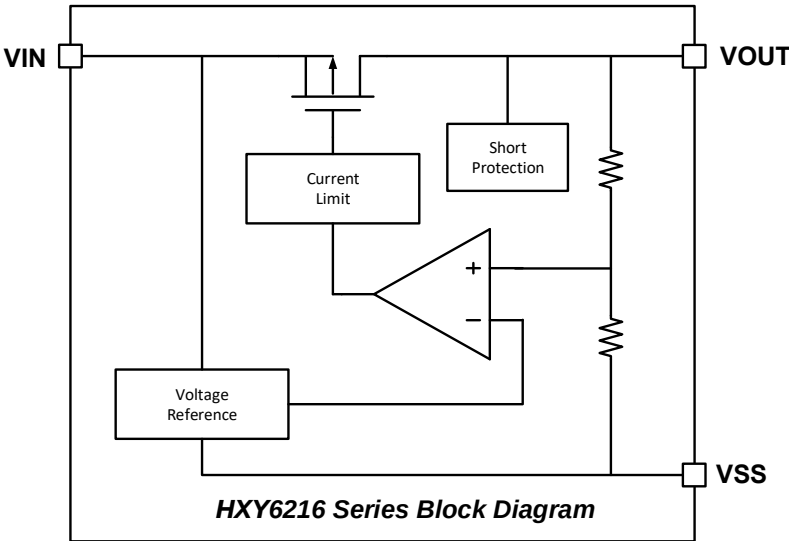
产品型号	封装规格	丝印	数量信息(PCS)
XC6206P332MR	SOT-23	662K	3000

典型应用





功能框图



极限参数

(注意：超过这些限制可能会损坏器件。长期暴露在绝对最大额定条件下会影响器件的可靠性。)

项目	符号	极限值	单位
输入电压	V_{IN}	8	V
输出电流	I_{OUT}	300	mA
输出电压	V_{OUT}	$V_{SS}-0.3\sim V_{IN}+0.3$	V
功 耗	P_d	0.20	W
工作温度	T_{opr}	$-40\sim +85$	$^{\circ}C$
存储温度	T_{stg}	$-55\sim +125$	$^{\circ}C$



电学参数

($V_{IN}=5V, T_A=25^\circ C$, 除特别指定)

项目	符号	条件	最小值	典型值	最大值	单位
输出电压	$V_{OUT(E)}$	$V_{IN}=V_{OUT(S)}+1.0V$, $I_{OUT}=1mA, \pm 2\%$	$V_{OUT(S)} \times 0.98$	$V_{OUT(S)}$	$V_{OUT(S)} \times 1.02$	V
最大输出电流	I_{OUT}	$V_{IN} \geq V_{OUT(S)}+1.0V$	300	—	—	mA
负载稳定度	ΔV_{OUT}	$V_{IN}=V_{OUT}+1V$ $1mA \leq I_{OUT} \leq 100mA$	—	25	—	mV
跌落电压	V_{drop}	$1.5V \leq V_{OUT(S)} \leq 2.5V$ $I_{OUT}=50mA$	—	0.20	0.28	V
		$2.6V \leq V_{OUT(S)} \leq 3.3V$ $I_{OUT}=50mA$	—	0.16	0.24	
		$3.4V \leq V_{OUT(S)} \leq 6.0V$ $I_{OUT}=50mA$	—	0.12	0.20	
输入稳定度	$\frac{\Delta V_{OUT1}}{\Delta V_{IN} \cdot V_{OUT}}$	$V_{OUT(S)}+0.5V \leq V_{IN} \leq 5.5V$ $I_{OUT}=1mA$	—	0.05	0.2	%/V
静态电流	I_{SS}	$V_{IN}=V_{OUT(S)}+1.0V$		3		μA
输入电压	V_{IN}		1.8		6	V
输出电压温度系数	$\frac{\Delta V_{OUT}}{\Delta T_{OPR} \cdot V_{OUT}}$	$I_{OUT}=40mA$ $-40^\circ C \leq T_{opr} \leq 85^\circ C$		100		ppm/ $^\circ C$
纹波抑制比	PSRR	$V_{IN}=[V_{OUT}+1]V$ $+1Vp-pAC$ $I_{OUT}=10mA, f=1kHz$		40		dB
短路电流	I_{SHORT}	$V_{IN}=V_{OUT}+1.5V$, $V_{OUT}=V_{SS}$		50		mA
过流保护电流	I_{LIMIT}			300	350	mA

注:

1. $V_{OUT(S)}$ = 规定输出电压
2. $V_{OUT(E)}$ = 有效输出电压 (即当 I_{OUT} 保持一定数值, $V_{IN}=V_{OUT}+1V$, 时的输出电压)
3. $V_{drop} = \{V_{IN1} (\text{注 } 5) - V_{OUT1} (\text{注 } 4)\}$
4. $V_{OUT1} = V_{OUT(E)} \times 98\%$
5. V_{IN1} = 逐渐减小输入电压, 当输出电压降为 $V_{OUT(E)}1$ 的98%时的输入电压。
6. Unless otherwise stated, $V_{IN} = V_{OUT(S)}+1.0V$



特征曲线

(3.3V output)

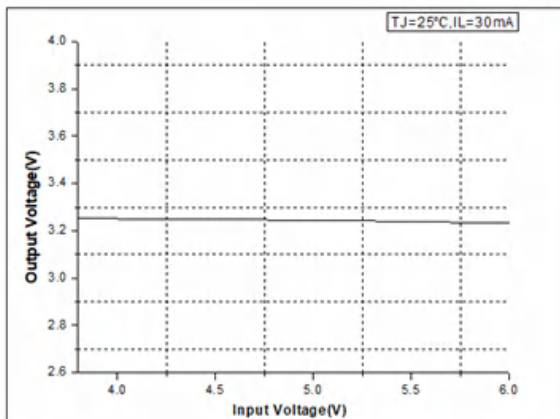


图1 输出电压和输入电压关系

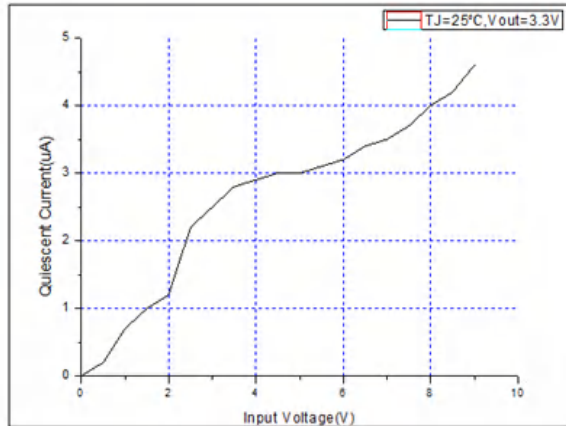


图2 静态功耗和输入电压关系

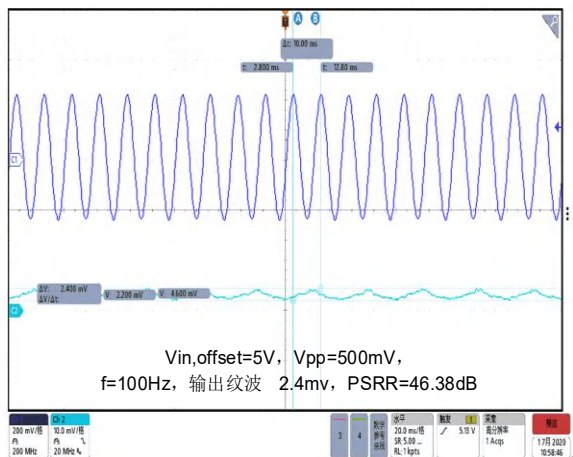


图3 纹波抑制比 (f=100Hz)

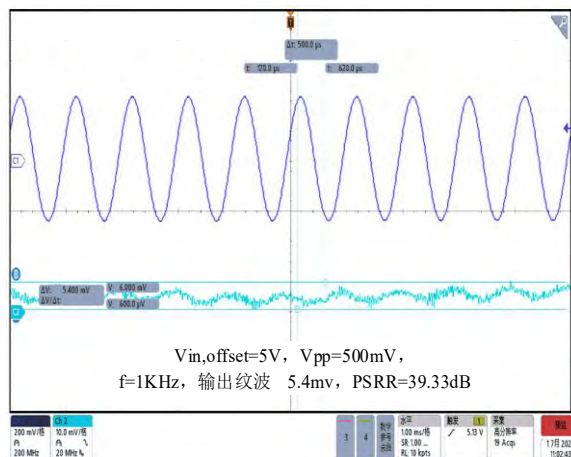


图4 纹波抑制比 (f=1KHz)

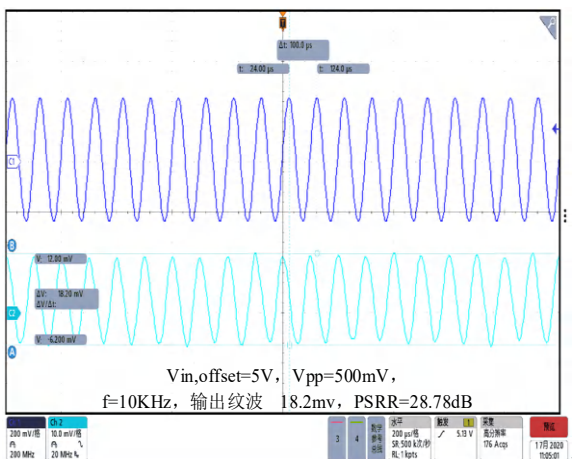


图5 纹波抑制比 (f=10Hz)

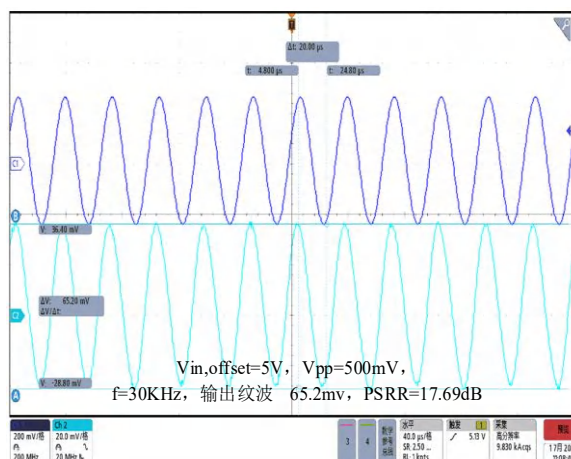
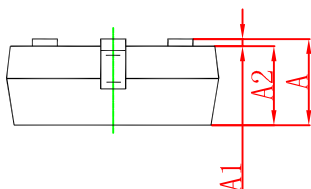
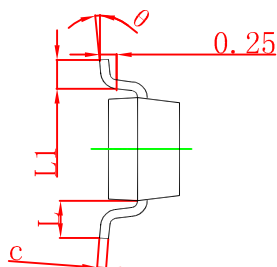
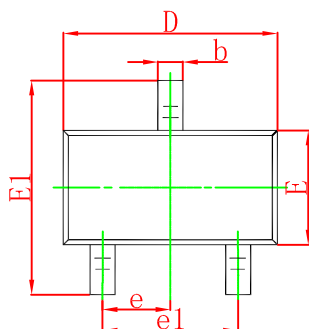


图6 纹波抑制比 (f=30Hz)

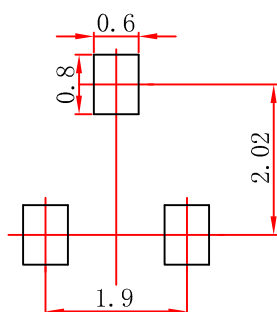


SOT-23 Package Outline Dimensions



Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min	Max	Min	Max
A	0.900	1.150	0.035	0.045
A1	0.000	0.100	0.000	0.004
A2	0.900	1.050	0.035	0.041
b	0.300	0.500	0.012	0.020
c	0.080	0.150	0.003	0.006
D	2.800	3.000	0.110	0.118
E	1.200	1.400	0.047	0.055
E1	2.250	2.550	0.089	0.100
e	0.950 TYP		0.037 TYP	
e1	1.800	2.000	0.071	0.079
L	0.550 REF		0.022 REF	
L1	0.300	0.500	0.012	0.020
θ	0°	8°	0°	8°

SOT-23 Suggested Pad Layout



Note:

1. Controlling dimension: in millimeters.
2. General tolerance: $\pm 0.05\text{mm}$.
3. The pad layout is for reference purposes only.



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